

5.3.5 Embedded voltage reference

The parameters provided in Table 22. [Embedded internal voltage reference](#) are derived from tests performed under the ambient temperature and supply voltage conditions summarized in [Section 5.3.1](#).

Table 22. Embedded internal voltage reference

The values in this table are specified by design and not tested in production.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V _{REFINT}	Internal reference voltages	-40 °C < T _J < 140 °C	1.180	1.217	1.250	V
t _{S_vrefint} ⁽¹⁾⁽²⁾	ADC sampling time when reading the internal reference voltage	-	4.3	-	-	μs
t _{start_vrefint} ⁽²⁾	Start time of reference voltage buffer when ADC is enable	-	-	-	4.4	μs
I _{refbuf} ⁽²⁾	Reference Buffer consumption for ADC	V _{DDA} = 3.3 V	9	13.5	23	μA
ΔV _{REFINT} ⁽²⁾	Internal reference voltage spread over the temperature range	-40 °C < T _J < 140 °C	-	5	15	mV
T _{coeff}	Average temperature coefficient	Average temperature coefficient	-	19	67	ppm/°C
V _{DDcoeff}	Average Voltage coefficient	3.0 V < V _{DD} < 3.6 V	-	10	1370	ppm/V

1. The shortest sampling time for the application can be determined by multiple iterations.

2. Specified by design - Not tested in production.

Table 23. Internal reference voltage calibration values

Symbol	Parameter	Memory address
V _{REFINT_CAL}	Raw data acquired at temperature of 30 °C, V _{DDA} = 3.3 V	0x08FFF810 - 0x08FFF811

5.3.6 Supply current characteristics

The current consumption is a function of several parameters and factors such as the operating voltage, ambient temperature, I/O pin loading, device software configuration, operating frequencies, I/O pin switching rate, program location in memory and executed binary code.

The current consumption is measured as described in [Current consumption measurement](#).

Typical and maximum current consumption

The MCU is placed under the following conditions:

- All I/O pins are in analog input mode.
- All peripherals are disabled except when explicitly mentioned.
- The flash memory access time is adjusted with the minimum wait-state number, depending on the f_{HCLK} frequency (refer to the tables *Number of wait states according to CPU clock (HCLK) frequency* available in the product reference manual).
- When the peripherals are enabled, f_{PCLK} = f_{HCLK}.

The parameters given in the tables below are derived from tests performed under ambient temperature and supply voltage conditions summarized in [Section 5.3.1: General operating conditions](#). If not specified otherwise, typical data are measured with a V_{DD} supply of 3.0 V, and maximum data are measured at 3.6 V.